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ELECTRODE FOR RECHARGEABLE BATTERY AND ELECTRODE ASSEMBLY

Abstract

An electrode for a rechargeable battery is provided. The electrode includes a substrate having an electrode uncoated region and an electrode active region. An electrode tab is attached to the electrode uncoated region, and an active material layer is formed on the electrode active region. A tape is disposed between the electrode uncoated region and the electrode tab.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and the benefit of Korean Patent Application No. 10-2024-0030049 filed in the Korean Intellectual Property Office on Feb. 29, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

(a) Field of the Invention

[0002] The present disclosure relates to an electrode assembly. More particularly, the present disclosure relates to an electrode for a rechargeable battery and an electrode assembly including the electrode.

(b) Description of the Related Art

[0003] A rechargeable battery is a power storage system that converts electrical energy into chemical energy to store it with excellent energy density. Unlike a primary battery that may not be recharged, the rechargeable battery may be recharged, and it may be widely used in information technology (IT) devices such as a smartphone, a laptop computer, a tablet computer, and the like.

[0004] Recently, there has been an increased interest in electric vehicles due to problems such as environment degradation and fossil fuel depletion, and rechargeable batteries may be used as the batteries for the electric vehicles.

[0005] Rechargeable batteries may be divided into sheet-shaped laminated batteries and wound jelly roll types. In the case of wound-type rechargeable batteries, cracks may occur due to curvature of the battery structure.

[0006] An electrode tab may be a relatively rigid material compared to an electrode substrate, and the smaller the curvature is, the more easily cracks occur around the electrode tab. Particularly, as the electrode assembly expands according to the charge and discharge of the rechargeable battery, cracks occur more easily as the electrode tab is pushed.

[0007] The above-described information disclosed in the technology behind this disclosure is only intended to improve understanding of the background of the present disclosure, and may therefore include information that does not constitute conventional art.

SUMMARY OF THE DISCLOSURE

[0008] The present disclosure provides an electrode for a rechargeable battery in which cracks are reduced when an electrode tab is attached to a position with small curvature, and an electrode assembly including the electrode.

[0009] An embodiment of the present disclosure provides an electrode for a rechargeable battery including: a substrate having an electrode uncoated region and an electrode active region; an electrode tab attached to the electrode uncoated region; an active material layer formed on the electrode active region; and a tape disposed between the electrode uncoated region and the electrode tab.

[0010] A width of the substrate exposed between the active material layer and the tape may be 1.5 mm to 5 mm.

[0011] A width of the substrate exposed between the active material layer and the tape may be 5% to 30% of the width of the electrode uncoated region.

[0012] The tape may overlap 40% to 85% of a width of the electrode uncoated region.

[0013] A thickness of the tape may be equal to or greater than 20% of a thickness of the active material layer.

[0014] The tape may overlap the electrode tab by 5% or more of an entire length of the electrode tab.

[0015] An entire length of the tape may be equal to or less than an entire width of the substrate.

[0016] The electrode may be a positive electrode.
[0017] The electrode tab may be made of aluminum.
[0018] The tape may be made of polyimide.
[0019] The tape may be a lower tape and the electrode may further include an upper tape for covering the electrode tab, with the upper tape having an end portion disposed on the active material layer.
[0020] The upper tape may be thicker than the lower tape.
[0021] The active material layer may be disposed on opposite sides of the electrode uncoated region.
[0022] The electrode may further include an upper tape for covering the electrode tab and having an end portion disposed in the electrode uncoated region.
[0023] The upper tape is shorter than the width of the substrate such that the lower tape is exposed.
[0024] Another embodiment of the present disclosure provides an electrode assembly including: a first electrode formed of the electrode for a rechargeable battery; a second electrode overlapping the first electrode; and a separation film disposed between the first electrode and the second electrode, wherein the first electrode, the separation film, and the second electrode are wound.
[0025] According to the present disclosure, the lower tape is attached, and the electrode tab is attached to the lower tape to reduce the generation of cracks around the electrode tab. Accordingly, a safe electrode is provided for a rechargeable battery and an electrode assembly including the same.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The following drawings appended to the specification illustrate embodiments of the present invention to serve a technical concept of the present disclosure to be understood easily together with detailed descriptions of the present invention, so the present invention should not be meant to restrict the disclosure to the matters illustrated in the accompanying drawings.
[0027] FIG. 1 shows an electrode according to an embodiment.
[0028] FIG. 2 shows a cross-sectional view with respect to a line II-II' of FIG. 1.
[0029] FIG. 3 and FIG. 4 show a spread electrode according to another embodiment.
[0030] FIG. 5 shows a cross-sectional view of a cylindrical rechargeable battery according to an embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0031] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. Terms or words used in the present specification and claims, which will be described below, should not be interpreted as being limited to typical or dictionary meanings, but should be interpreted as having meanings and concepts which comply with the technical spirit of the present disclosure, based on the principle that an inventor can appropriately define the concept of the terms to describe his/her own invention in the best manner. Therefore, configurations illustrated in the embodiments and the drawings described in the present specification are only the most preferred embodiment of the present disclosure and do not represent all of the technical spirit of the present disclosure, and thus it is to be understood that various equivalents and modified examples, which may replace the configurations, are possible when filing the present application.
[0032] When used in the present specification, “comprise and include” and/or “comprising and including” may specify the existence of the mentioned shapes, numbers, steps, operations, members, elements and/or these groups, and may not exclude the presence or addition of one or more other shapes, numbers, movements, members, elements and/or groups.

[0033] To aid understanding of the disclosure, the attached drawings may not be drawn to actual scale, but the dimensions of some components may be exaggerated. The same reference number may be assigned to the same component in another embodiment.

[0034] Although first, second, etc. are used to describe various components, the components may not be limited by these terms. These terms are only used to distinguish one component from another, and unless specifically stated to the contrary, the first component may also be the second component.

[0035] Throughout the specification, unless otherwise stated, each component may be singular or plural.

[0036] As shown in the drawings, to describe the relationship between one element or feature and other element(s) or feature(s), for ease of description, spatial relative terms such as beneath, below, lower, above, upper, etc. may be used in the present specification. Spatially relative dispositions will be understood to encompass different directions of the device in use or operation in addition to the direction depicted in the figures. For example, if the drawing device is flipped, an element described as “below” or “bottom” another element may be understood to be “above” or “beyond” another element. Therefore, the term “down” may encompass both up and down directions.

[0037] It should be understood that if a component is described as “connected to” or “coupled to” another component, the components may be directly connected or accessed to each other, but other components may be “interposed” between the respective components, or the respective components may be connected, combined, or accessed to each other through other components.

[0038] The terms used in this specification are intended to describe embodiments of the present disclosure and are not intended to limit the present disclosure.

[0039] The rechargeable battery according to an embodiment will now be described in detail with reference to the accompanying drawings.

[0040] FIG. 1 shows an electrode according to an embodiment, and FIG. 2 shows a cross-sectional view with respect to a line II-II' of FIG. 1.

[0041] As shown in FIG. 1 and FIG. 2, a first electrode **131** may include a first substrate **31**, a first electrode active region DA including a first active material layer **33** formed on a first substrate **31**, and a first electrode uncoated region DB in which the first active material layer **33** is not formed and the first substrate **31** is exposed.

[0042] A first electrode tab **135** may be connected to the first electrode uncoated region DB. The first electrode tab **135** may be made of a same material as the substrate, e.g., aluminum.

[0043] The disposition of the electrode tab may be variable according to resistance and output of the electrode assembly, and as shown, the first electrode active region DA may be formed to be separated into a right portion and a left portion with respect to the first electrode tab. The first substrate **31** exposed between a separated gap may be the first electrode uncoated region DB.

[0044] A lower tape **81** may be attached to a portion below the first electrode tab **135**. For example, a lower tape **81** may be disposed between a first surface A of the first substrate **31** and the first electrode tab **135**, and the lower tape **81** may be attached to the first surface A of the substrate **31**. The lower tape **81** may be made of polyimide. The lower tape **81** may reduce a step between the first electrode tab **135** and the first active material layer **33** due to a thickness of the lower tape **81**. If the step between the first electrode tab **135** and the first active material layer **33** is significant, cracks may be caused by a volume expansion at the time of charge and discharge. An embodiment of the present disclosure may reduce the step and may accordingly reduce the generation of cracks when the volume expands because of the charge and discharge.

[0045] The first electrode tab **135** may be manufactured by a slitting process, thereby forming burrs on an edge of the first electrode tab **135**. The burrs may protrude to damage the first substrate **31**. In an embodiment of the present disclosure, the lower tape **81** prevents the burrs on the edge of the first electrode tab **135** from contacting the first substrate **31**. That is, if pressure is applied to the burrs, the lower tape **81** absorbs the pressure to thus prevent the burrs from piercing and damaging

the first substrate **31**.

[0046] The lower tape **81** may be formed to be larger than the first electrode tab **135** and may extend so as to be higher than a protruding distance of the burrs. A width **D1** of the lower tape **81** may be equal to or greater than twice a width **D2** of the first electrode tab **135**, and may be 40% to 85% of a width **D3** of the first electrode uncoated region **DB**. A gap **D4** between the lower tape **81** and first active material layer **33** may be 1.5 mm to 5 mm, and may be 5% to 30% of the width **D3** of the first electrode uncoated region **DB**. A thickness **T1** of the lower tape **81** may be 20% a thickness **T2** of the first active material layer **33**. Therefore, the step between the lower tape **81** and the first active material layer **33** may be equal to or less than 80% of the thickness **T2** of the first active material layer **33**. If the thickness **T1** of the lower tape **81** is greater, the first electrode tab **135** may protrude from the first active material layer **33** and, thus, may not be easily wound as a result of the thickness **T1** of the lower tape **81**.

[0047] An entire length of the lower tape **81** may be equal to or less than an entire width **D5** of the first substrate **31** and may overlap a portion of the first electrode tab **135**. A length **L2** by which the lower tape **81** overlaps the first electrode tab **135** may be equal to or greater than 5% of the entire length **L3** of the first electrode tab **135**.

[0048] An upper tape **82** may be attached to the first electrode tab **135**. The upper tape **82** may cover the first electrode tab **135** and may protect the first electrode tab **135**. An end portion of the upper tape **82** may be disposed on the neighboring first active material layer **33**. The upper tape **82** may be attached to one or more parts of the lower tape **81** that are exposed between the first active material layer **33** and the first electrode tab **135** or on an upper surface of the first substrate **31**. The upper tape **82** may more firmly attach the first electrode tab **135** to the first substrate **31**. The upper tape **82** is shown to cover the electrode uncoated region and an end portion of the active material layer **33**, but is not limited thereto.

[0049] The thickness of the upper tape may be greater than the thickness **T1** of the lower tape **81**, for example, the thickness of the upper tape **82** may be 20 μm to 50 μm , preferably 30 μm , and the thickness of the lower tape **81** may be 10 μm to 20 μm , preferably 18 μm .

[0050] FIG. 3 and FIG. 4 show a spread electrode according to another embodiment.

[0051] As shown in FIG. 3 and FIG. 4, the upper tape **82** may be equal to or less than the width **D3** of the electrode uncoated region **DB**. Therefore, the end portion of the active material layer **33** disposed on border of the electrode uncoated region **DB** may not be covered by the upper tape **82** and may be exposed. That is, the active material layer **33** may be spaced from the border of the upper tape **82**.

[0052] As shown in FIG. 3, the upper tape **82** may be formed to have a greater length than the width of the substrate **31**. Without being limited thereto, as shown in FIG. 4, the upper tape **82** may cover a portion of the electrode tab **135** and the lower tape **81**. In particular, the upper tape **82** may cover the end portion of the electrode tab **135**, and the lower tape **81** may be partly exposed.

[0053] The first electrode of FIG. 1 and FIG. 2 may be used as the first electrode or the second electrode of the cylindrical rechargeable battery. This will now be described with reference to FIG. 5.

[0054] FIG. 5 shows a cross-sectional view of a cylindrical rechargeable battery according to an embodiment.

[0055] As shown in FIG. 5, the rechargeable battery **100** according to an embodiment may include a case **120**, an electrode assembly **130** received in the case **120**, and a cap assembly **140** assembled in an opening of the case **120** and sealing the case **120**. The cap assembly **140** may include a safety vent **10** for preventing explosion of the rechargeable battery **100**, and an upper cap **40** for covering the safety vent **10**.

[0056] The electrode assembly **130** may include a first electrode **131**, a separation film **133**, and a second electrode **132** that are sequentially stacked. The electrode assembly **10** may be a cylindrical jelly roll generated by stacking the first electrode **131**, the separation film **133**, and the second

electrode **132** and winding the same.

[0057] The first electrode **131** may be the electrode shown in FIG. **1** and FIG. **2**, may include a first substrate and a first active material layer formed on the first substrate, and the first electrode **131** may be a positive electrode. The first substrate may include a first electrode uncoated region and a first electrode active region, and a positive electrode tab that is the first electrode tab **135** may be attached to the first electrode uncoated region.

[0058] The first substrate may be formed of a thin conductive metal plate and may be used as a current collector, for example, it may be aluminum.

[0059] A compound (lithiated intercalation compound) for allowing a reversible intercalation and deintercalation of lithium may be used as the positive active material forming the first active material layer. In detail, at least one of composite oxides of a metal selected from cobalt, manganese, nickel, aluminum, and combinations thereof, and lithium may be used. A content of the positive active material may be 90 wt % to 98 wt % of the entire weight of the positive active material layer.

[0060] The positive active material may further include a binder and a conductive material. The contents of the binder and the conductive material may respectively be 1 wt % to 5 wt % of the entire weight of the positive active material layer.

[0061] The binder may attach the positive active material particles to each other, and may attach the positive active material to the substrate that is a current collector. For example, the binder may use polyvinylalcohol, carboxylmethylcellulose, hydroxypropylcellulose, deacetylcellulose, polyvinylchloride, carboxylated polyvinyl chloride, polyvinylfluoride, ethylene oxide-containing polymer, polyvinylpyrrolidone, polyurethane, polytetrafluoroethylene, polyvinylidene fluoride, polyethylene, polypropylene, styrene butadiene rubber, acrylated styrene butadiene rubber, epoxy resin, and nylon, but is not limited thereto. The conductive material may provide conductive to the electrode, and any materials may be usable if they generate no chemical changes and they may be electron conductive materials, regarding the configured batteries.

[0062] The second electrode **132** may include a second substrate and a second active material layer formed on the second substrate. The second substrate may include a second electrode active region in which a second active material layer is formed, and a second electrode uncoated region in which the active material layer is not formed and the second substrate is exposed. A negative electrode tab that is a second electrode tab **136** may be connected to the second electrode uncoated region. The second electrode tab **136** may be made of nickel.

[0063] As shown in FIG. **1** and FIG. **2**, regarding the second electrode, a lower tape may be attached to the second electrode uncoated region, and the second electrode tab **136** may overlap the lower tape. The second electrode **132** may be a negative electrode, and the second substrate may be made of a thin conductive metal plate and may be used as the current collector. For example, the second substrate **33** may be made of copper (Cu).

[0064] The negative active material of the negative active material layer may be a carbon-based active material. The carbon-based negative active material may be artificial graphite or a mixture of artificial graphite and natural graphite. If the artificial graphite or a crystalline carbon-based material that is the mixture of artificial graphite and natural graphite is used as the negative active material, a crystalline characteristic of particles may be further developed compared to the case of using an amorphous carbon-based active material, thereby further increasing an alignment characteristic of the carbon material in a polar plate on an external magnetic field, which is a merit. Shapes of the artificial graphite or the natural graphite may include an amorphous shape, a plate shape, a flake shape, a spherical shape, a fiber shape, and combinations thereof, and any types of other shapes. If the artificial graphite and the natural graphite are mixed and used, its mixture ratio may be 50:50 wt % to 95:5 wt %.

[0065] The negative active material layer may further include at least one of an Si-based negative active material, an Sn-based negative active material, and an LIMO_x (M is a metal)-based negative

active material. If the negative active material layer further includes them, for example, if it includes the carbon-based negative active material as the first negative active material and the negative active material as the second negative active material, the mixture ratio of the first negative active material to the second negative active material may be 50:50 to 99:1 weight ratio.

[0066] The LiMO_x (M is a metal)-based negative active material may be a lithium vanadium oxide.

[0067] The Si-based negative active material may include Si, a Si—C composite, a SiO₂.sub.x (0<x<2), and a Si-Q alloy (the Q is selected from among an alkali metal, an alkaline-earth metal, a Group 13 element, a Group 14 element, a Group 15 element, a Group 16 element, a transition metal, a rare earth element, and combinations thereof, and Si is not selected therefrom), and the Sn-based negative active material may include Sn, a SnO₂.sub.2, and a Sn—R alloy (the R is selected from among an alkali metal, an alkaline-earth metal, a Group 13 element, a Group 14 element, a Group 15 element, a Group 16 element, a transition metal, a rare earth element, and combinations thereof, and Sn is not selected therefrom), and at least one of them and the SiO₂.sub.2 may be mixed and used. The elements Q and R may be selected from among Mg, Ca, Sr, Ba, Ra, Sc, Y, Ti, Zr, Hf, Rf, V, Nb, Ta, Db, Cr, Mo, W, Sg, Tc, Re, Bh, Fe, Pb, Ru, Os, Hs, Rh, Ir, Pd, Pt, Cu, Ag, Au, Zn, Cd, B, Al, Ga, Sn, In, Tl, Ge, P, As, Sb, Bi, S, Se, Te, Po, and combinations thereof.

[0068] The content of the negative active material on the negative active material layer may be 95 wt % to 99 wt % of the entire weight of the negative active material layer.

[0069] The negative active material may include a binder and may further optionally include a conductive material. The content of the binder in the negative active material may be 1 wt % to 5 wt % of the entire weight of the negative active material. If the conductive material is further included, 90 wt % to 98 wt % of the negative active material, 1 wt % to 5 wt % of the binder, and 1 wt % to 5 wt % of the conductive material may be used.

[0070] The binder may attach the negative active material particles to each other and may attach the negative active material to the negative electrode substrate. A nonaqueous binder, an aqueous binder, or a combination thereof may be used as the binder.

[0071] The nonaqueous binder may include polyvinylchloride, carboxylated polyvinylchloride, polyvinylfluoride, an ethylene oxide-containing polymer, polyvinylpyrrolidone, polyurethane, polytetrafluoroethylene, polyvinylidene fluoride, polyethylene, polypropylene, polyamideimide, polyimide, and combinations thereof.

[0072] The aqueous binder may include styrene-butadiene rubber, acrylated styrene-butadiene rubber (SBR), acrylonitrile-butadiene rubber, acryl rubber, butyl rubber, ethylenepropylenecopolymer, polyepichlorohydrin, polyphosphazene, polyacrylonitrile, polystyrene, ethylenepropylene diene copolymer, polyvinylpyridine, chlorosulfonated polyethylene, latex, polyester resin, acryl resin, phenol resin, epoxy resin, polyvinyl alcohol, acrylate-based resin, and combinations thereof.

[0073] If the aqueous binder is used as the negative electrode binder, a cellulose-based compound for providing viscosity may be further included as a thickener. At least one of carboxymethyl cellulose, hydroxypropylmethyl cellulose, methyl cellulose, or alkali metal salt thereof may be mixed and used as the cellulose-based compound. Na, K or Li may be used as the alkali metal. The content of using the thickener may be 0.1 parts by weight to 3 parts by weight with respect to the negative active material 100 parts by weight.

[0074] The conductive material is used to provide conductivity to the electrode, and any types of electron conductive materials generating no chemical changes are usable for the configured battery. Examples of the conductive material may include carbon-based materials including natural graphite, artificial graphite, carbon black, acetylene black, Ketjen black, and carbon fiber; metal-based materials such as metal powder or metal fiber including copper, nickel, aluminum, and silver; conductive polymers including a polyphenylene derivative; and conductive materials including mixtures thereof.

[0075] A Brunauer-Emmett-Teller (BET) specific surface area of the negative active material layer may be less than 3.0 m²/g, and may be 0.6 m²/g to 1.2 m²/g. It may be advantageous

that the BET specific surface area of the negative active material layer is less than 3.0 m²/g because an electrochemical lifespan characteristic of cells may then be increased.

[0076] The BET may be measured by charging and discharging the lithium rechargeable battery including the negative electrode, disassembling the completely discharged battery to obtain a negative electrode, cutting the negative electrode by a predetermined size, putting the cut negative electrodes into a BET sample holder, and applying a nitrogen gas adsorption method thereto.

[0077] The negative electrode may have a cross-section loading level (L/L) of 6 mg/cm² to 65 mg/cm².

[0078] The separation film **133** may be disposed between the first electrode **131** and the second electrode **132** and may insulate them. Polyethylene, polypropylene, polyvinylidene fluoride, or a multilayer with at least two layers thereof may be used as the separation film **133**, and mixed multilayers such as a two-layered separation film of polyethylene/polypropylene, a three-layered separation film of polyethylene/polypropylene/polyethylene, or a three-layered separation film of polypropylene/polyethylene/polypropylene may also be used.

[0079] The electrode assembly **130** may be wound with respect to a center pin **134** so the center pin **134** may be disposed in a center of the electrode assembly **130** and may be arranged in parallel to a direction in which the electrode assembly **130** is inserted into the case **120**.

[0080] The center pin **134** having a circular pipe shape with an empty inside may allow a minimum transformation or may maintain the shape that is close to the shape prior to transformation if receiving a total compressing weight or a partial impact weight applied from the outside of the rechargeable battery. The center pin **134** may function as a passage for moving gas generated in an inside. The center pin **134** may be omitted if needed.

[0081] The center pin **134** may be made of a material with rigidity, for example, a conductive metal such as steel, a steel alloy, aluminum, or an aluminum alloy, so that it may be transformed to the minimum for the external impact. The center pin **134** may be conductive, so a first insulation plate **137** may be disposed between the upper cap **40** and the center pin **134** and a second insulation plate **138** may be disposed between a bottom portion **121** of the case **120** and the center pin **134** so that respective ends of the center pin **134** may be maintained to be insulated.

[0082] A penetration hole that the center pin **134** passes through, a penetration hole that the first electrode tab **135** passes through, and penetration holes through which an electrolyte is input may be formed in the first insulation plate **137**. A penetration hole for the center pin **134** and a penetration hole for the second electrode tab **136** may be formed in the second insulation plate **138**.

[0083] One side of the case **120** may be opened and the case **120** may have substantially the same shape as the electrode assembly **130** such that the electrolyte and the electrode assembly **130** may be inserted. The case **120** may include a circular bottom portion and a cylindrical lateral portion extending upward from the bottom portion by a predetermined length. An upper portion of the cylindrical case may be opened while the rechargeable battery is assembled.

[0084] Therefore, the electrode assembly may be inserted into the cylindrical case during the process for assembling a rechargeable battery, and the electrolyte solution may be injected into the cylindrical case. One side (upper side) of the case **120** may be opened during the process for manufacturing a rechargeable battery **100**, and the electrode assembly **130** and the electrolyte may be received in the case **120**. The case **120** may be made of steel, a steel alloy, aluminum, and an aluminum alloy.

[0085] The electrolyte may allow lithium ions generated by an electrochemical reaction to move in the first electrode and the second electrode in the battery. The electrolyte solution may be made of lithium salt such as LiPF₆ or LiBF₄ in an organic solvent such as ethylene carbonate (EC), propylene carbonate (PC), dimethyl carbonate (DEC), or ethyl methyl carbonate (EMC), or dimethyl carbonate (DMC). The electrolyte may be liquid, solid, or gel.

[0086] A beading portion **123** and a crimping portion **124** may be disposed on a lateral portion **122** of the case **120**.

[0087] The beading portion **123** may be transformed to be concave toward the inside of the case **120**, and the crimping portion **124** may be transformed so that an edge of the lateral portion **122** may be bent toward the inside. A movement of the electrode assembly **130** may be suppressed by the beading portion **123**, and the cap assembly **140** may be fixed to the case **120** by the crimping portion **124**.

[0088] The cap assembly **140** may include a safety vent **10** for preventing an explosion of the rechargeable battery **100**, and an upper cap **40** for covering the safety vent **10**.

[0089] The cap assembly **140** may include a safety vent **10** including a notch groove, a cap-down **20** disposed on one side (lower side) of the safety vent **10** facing the electrode assembly **130**, a ring-type insulator **30** disposed between the safety vent **10** and the cap-down **20**, and an upper cap **40** disposed on one side (upper side) of the safety vent **10** that is opposite the cap-down **20**. The safety vent **10** may be referred to as a current interruptive device (CID).

[0090] The first electrode tab **135** of the electrode assembly **130** may be fixed to a first surface (lower surface) of the lower cap **20**, and the lower cap **20**, the safety vent **10**, and the upper cap **40** may be charged with the positive electrode. The first electrode tab **135** may be bent to face the first surface of the lower cap **20** to thereby increase a contact area with the lower cap **20**.

[0091] The upper cap **40** may protrude outside of the rechargeable battery **100** to function as a positive electrode terminal for contacting an external device and flowing the current, and the upper cap **40** may have a planar surface.

[0092] The second electrode tab **136** may be connected to the second electrode uncoated region and may be fixed to a lower bottom surface of the case **120** by welding. Thus, the second electrode tab **136** may protrude in an opposite direction to the first electrode tab **135**. Therefore, the case **120** may be charged with the negative electrode, and the bottom portion **121** of the case **120** may function as the negative terminal.

[0093] The cap assembly **140** may be combined to the lateral portion **122** of the case **120** with the insulation gasket **141** as a medium. The insulation gasket **141** may surround edges of the safety vent **10** and the upper cap **40**. The insulation gasket **141** may be compressed between the beading portion **123** and the crimping portion **124** of the case **120**.

[0094] While the rechargeable battery **100** is used, gas may be generated in the case **120** because of many factors. And an internal pressure of the rechargeable battery **100** may be increased by the gas. If the gas is generated, a pressure may be continuously applied to the safety vent **10** through an opening of the lower cap **20**, the safety vent **10** may be transformed toward the external side (or upper side) of the cap assembly **140** at a specific pressure, and the safety vent **10** may be separated from the lower cap **20**.

[0095] The current flow may be prevented by the separation of the safety vent **10** and the lower cap **20**. If the pressure is continuously increased, the notch of the safety vent **10** may be fractured and the internal gas may be discharged. The internal gas may be discharged to outside of the rechargeable battery **100** through an exhaust hole formed in the upper cap **40**.

[0096] Table 1 shows data generated by measuring cracks according to an embodiment of the present disclosure and a comparative example.

TABLE-US-00001 Covered Distance D4 amount D1 between tape of uncoated and active region in material width Horizontal Vertical layer direction cracks cracks Embodiment 1 5 mm 44% Not Not generated generated Embodiment 2 1.5 mm 83% Not Not generated generated Comparative -2.5 mm 100% Generated Not Example generated

In particular, Table 1 shows the measured generation of cracks for Embodiments 1 and 2 and the comparative example in the cylindrical electrode assembly of the rechargeable battery shown in FIG. 3 to FIG. 5. Regarding the cracks of the active material layer, vertical cracks that are in the length direction of the lower tape and horizontal cracks that are in the width direction of the lower tape may be generated according to the motion so the generation of cracks according to the respective directions may be measured. In the embodiments 1 and 2, the gaps between the lower

tape and the active material layer were, respectively 5 mm and 1.5 mm, and the lower tape overlapped the electrode uncoated region by the ratios of 44% and 83% of the width of the electrode uncoated region. In the comparative example, there was no gap between the lower tape and the active material layer, and the lower tape overlapped the neighboring active material layer by 2.5 mm. Therefore, the entire (100%) width of the electrode uncoated region was covered with the lower tape.

[0097] Referring to Table 1, it can be seen that the vertical cracks were not generated in the embodiment and the comparative example, the horizontal cracks were not generated if the gap between the lower tape and the active material layer is maintained as in the embodiments 1 and 2, and the horizontal cracks are generated if there is no such gap as in the comparative example.

[0098] While this disclosure has been described in connection with what is presently considered to be practical embodiments, it is to be understood that the disclosure is not limited to the disclosed embodiments. On the contrary, is intended to cover various modifications and equivalent arrangements included within the spirit and scope of the appended claims.

TABLE-US-00002 <Description of symbols> 100: rechargeable battery 120: case 130: electrode assembly 131: positive electrode 132: negative electrode 133: separation film 135: first electrode tab 136: second electrode tab

Claims

1. An electrode for a rechargeable battery, the electrode comprising: a substrate having an electrode uncoated region and an electrode active region; an electrode tab attached to the electrode uncoated region; an active material layer formed on the electrode active region; and a tape disposed between the electrode uncoated region and the electrode tab.
2. The electrode as claimed in claim 1, wherein a width of the substrate exposed between the active material layer and the tape is 1.5 mm to 5 mm.
3. The electrode as claimed in claim 1, wherein a width of the substrate exposed between the active material layer and the tape is 5% to 30% of the width of the electrode uncoated region.
4. The electrode as claimed in claim 1, wherein the tape overlaps 40% to 85% of a width of the electrode uncoated region.
5. The electrode as claimed in claim 1, wherein a thickness of the tape is equal to or greater than 20% of a thickness of the active material layer.
6. The electrode as claimed in claim 1, wherein the tape overlaps the electrode tab by 5% or more of an entire length of the electrode tab.
7. The electrode as claimed in claim 1, wherein an entire length of the tape is equal to or less than an entire width of the substrate.
8. The electrode as claimed in claim 1, wherein the electrode tab is made of aluminum.
9. The electrode as claimed in claim 1, wherein the tape is made of polyimide.
10. The electrode as claimed in claim 1, wherein the tape is a lower and tape, and the electrode further comprises an upper tape for covering the electrode tab, with the upper tape having an end portion disposed on the active material layer.
11. The electrode as claimed in claim 10, wherein the upper tape is thicker than the lower tape.
12. The electrode as claimed in claim 10, wherein the active material layer is disposed on opposite sides of the electrode uncoated region.
13. The electrode as claimed in claim 10, wherein the tape is a lower and tape, and the electrode, and the electrode further comprises an upper tape for covering the first electrode tab and having an end portion disposed in the first electrode uncoated region.
14. The electrode as claimed in claim 10, wherein the upper tape is shorter than the width of the substrate such that the lower tape is exposed.
15. An electrode assembly comprising: first electrode including (i) a substrate having an electrode

uncoated region and an electrode active region, (ii) an electrode tab attached to the electrode uncoated region, (iii) an active material layer formed on the electrode active region, and (iv) a tape disposed between the electrode uncoated region and the electrode tab; a second electrode overlapping the first electrode; and a separation film disposed between the first electrode and the second electrode, wherein the first electrode, the separation film, and the second electrode are wound.
